IIITB COMET FOUNDATION

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Gate -PH 2011 46 Question

QUESTION

PH

The following Boolean expression

$Y = A \bullet \overline{B} \bullet \overline{C} \bullet \overline{D} + \overline{A} \bullet B \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet C \bullet D + \overline{A} \bullet B \bullet C \bullet D + A \bullet \overline{B} \bullet \overline{C} \bullet D$

can be simplified to

(A) $\overline{A} \bullet \overline{B} \bullet C + A \bullet \overline{D}$

(B) $\overline{A} \bullet B \bullet \overline{C} + A \bullet \overline{D}$

(C) $A \bullet \overline{B} \bullet \overline{C} + \overline{A} \bullet D$

(D) $A \bullet \overline{B} \bullet C + \overline{A} \bullet D$

QUESTION ANALYSIS

Solution

can be simplified to

(A) $\overline{A} \cdot \overline{B} \cdot C + A \cdot \overline{D}$ **(B)** $\overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{D}$

(C) $A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot D$ **(D)** $A \cdot \overline{B} \cdot C + \overline{A} \cdot D$

Solution

Given,

$$Y = A\overline{BCD} + \overline{A}B\overline{C}D + \overline{ABC}D + \overline{A}BCD + \overline{A}BCD + A\overline{BC}D$$

Step 1: Group the terms

Group the first and last terms:

$$A\overline{BCD} + A\overline{BC}D = A\overline{BC}(\overline{D} + D)$$

Since $\overline{D} + D = 1$,

$$A\overline{BC}(\overline{D} + D) = A\overline{BC}.$$

Now group the remaining four terms:

$$\overline{A}B\overline{C}D + \overline{ABC}D + \overline{A}BCD + \overline{A}BCD$$

$$= \overline{A}D\left(B\overline{C} + \overline{BC} + \overline{B}C + BC\right).$$

Step 2: Simplify inside the brackets

$$B\overline{C} + \overline{B\overline{C}} + \overline{B}C + BC$$

$$= \overline{C}(B + \overline{B}) + C(B + \overline{B})$$

$$= \overline{C}(1) + C(1)$$

$$= \overline{C} + C$$

$$= 1.$$

Therefore,

$$\overline{A}D(1) = \overline{A}D.$$

Step 3: Final expression

Hence,

$$\boxed{Y = A\overline{B\overline{C}} + \overline{A}D}$$

This matches **Option (C).**

$$\boxed{\text{Correct Answer: (C)}}$$

Hardware Components Required

S.No.	Component
1	ESP32-WROOM Development Board
2	Personal Computer (Ubuntu OS)
3	USB Cable
4	Breadboard
5	7447 BCD-to-7-Segment Decoder IC
6	Common Anode 7-Segment Display
7	220 Ω Resistors (7 Nos.)
8	Jumper Wires
9	5 V Power Supply (from ESP32 VIN/USB)

Table 1: Hardware and Software Components Required

S.No.	ESP32-WROOM Pin	7447 IC Pin
1	GPIO16	A
2	GPIO17	B
3	GPIO18	C
4	GPIO19	D
5	VIN (5V)	VCC (Pin 16)
6	GND	GND (Pin 8)

Table 2: ESP32-WROOM to 7447 IC Connections

S.No.	7447 Output	7-Segment Display
1	a	Segment a
2	b	Segment b
3	c	Segment c
4	d	Segment d
5	e	Segment e
6	f	Segment f
7	g	Segment g
8	Common Anode	+5 V
9	Segment Pins	Through 220 Ω Resistors

Table 3: 7447 IC to Common Anode 7-Segment Display Connections

Hardware Connections

Table 1: ESP32-WROOM to 7447 IC Connections

Table 2: 7447 IC to 7-Segment Display Connections

Table 3: Programming Connections

Execution Procedure

1. Create a new PlatformIO project for the ESP32-WROOM board.

   ```
   /root/venv/bin/pio project init --board esp32dev
   ```

2. Place the source code (`main.cpp`) inside the `src` directory of the project.

3. Compile the source code and generate the binary files using:

   ```
   /root/venv/bin/pio run
   ```

4. After successful compilation, the following binary files are generated in the build directory:

   `.pio/build/esp32dev/`

   - `bootloader.bin`

S.No.	Component	Connection
1	ESP32 USB Port	Ubuntu PC USB Port
2	USB Cable	Power and Program Upload
3	Ubuntu PC	PlatformIO and ESPTool

Table 4: Programming Connections

- `partitions.bin`
- `firmware.bin`

5. Copy the generated binary files to the desired directory on the Ubuntu PC.
6. Connect the ESP32-WROOM board to the Ubuntu PC using a USB cable.
7. Verify that the ESP32 board is detected by the operating system.

```
ls -l /dev/ttyUSB*
```

8. Verify the ESP32 chip.

```
esptool.py --chip esp32 --port /dev/ttyUSB0 chip_id
```

9. Flash the generated binary files into the ESP32.

```
esptool.py --chip esp32 \
  --port /dev/ttyUSB0 \
  --baud 460800 \
  write_flash \
  0x1000 bootloader.bin \
  0x8000 partitions.bin \
  0x10000 firmware.bin
```

10. Wait until the flashing process completes successfully.
11. Press the **EN (Reset)** button on the ESP32 board if required.
12. Observe the LED connected to the output GPIO pin to verify the implemented Boolean expression.

Results and Observations

The simplified Boolean expression was successfully implemented on the ESP32-WROOM development board. The output generated by the ESP32 was applied to the 7447 BCD-to-7-segment decoder IC, which drove the common anode 7-segment display. The displayed output was observed for all input combinations and found to be in agreement with the expected truth table.

Figure 1 shows the experimental output when the Boolean expression evaluates to logic LOW ($Y = 0$). In this case, the 7-segment display indicates the digit **0**.

Figure 2 shows the experimental output when the Boolean expression evaluates to logic HIGH ($Y = 1$). In this case, the 7-segment display indicates the digit **1**.

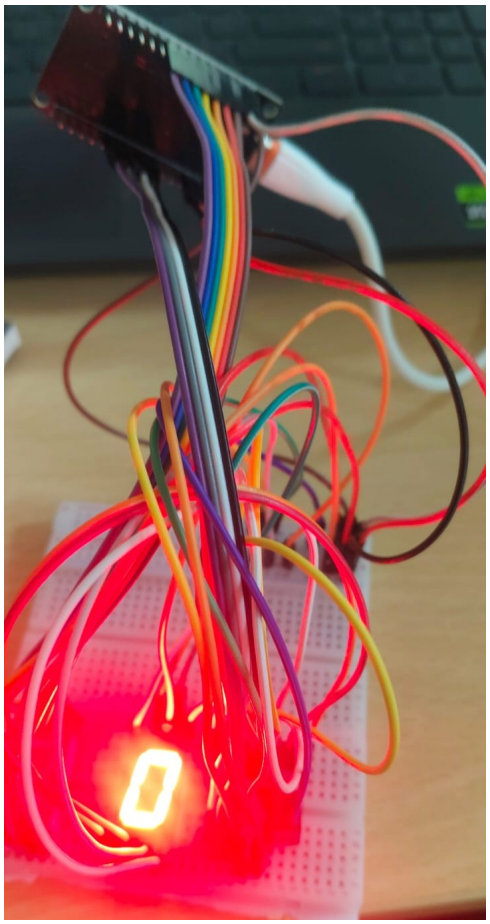


Figure 1: 7-Segment Display showing the digit '0' corresponding to logic LOW ($Y = 0$).

Result

The Boolean expression was successfully implemented and verified using the ESP32-WROOM development board, 7447 BCD-to-7-segment decoder IC, and common anode 7-segment display. The displayed outputs matched the expected results for all input combinations, thereby validating the correctness of the implemented Boolean function.

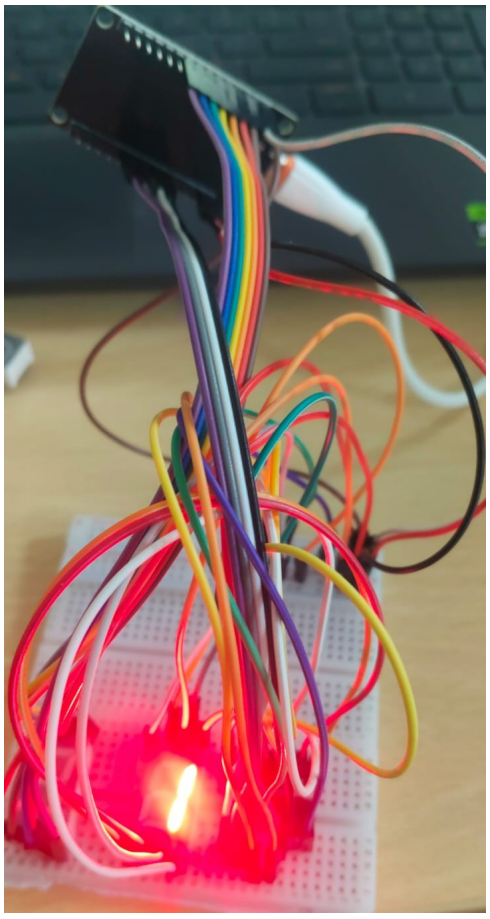


Figure 2: 7-Segment Display showing the digit '1' corresponding to logic HIGH ($Y = 1$).